

Quiz 2

1. [单选] Greedy Descent (0.5分)

Which of the following best characterizes the behavior of greedy descent (also known as iterative best improvement) in a discrete search space?

- ☐ It always finds the global optimum regardless of the starting point.
- ☐ It explores all neighboring solutions simultaneously before moving.
- ☒ It finds at a local optimum, which may not be globally optimal.
- ☐ It occasionally moves from a candidate solution to worse solutions.

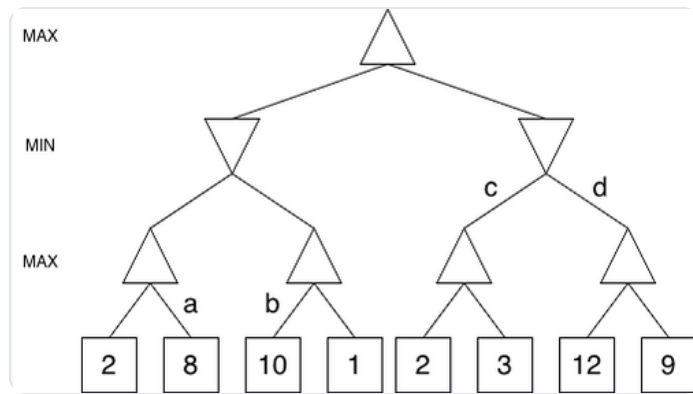
2. [单选] Gradient Descent (0.5分)

Consider the loss function $f(x) = x^2 + 4x$, starting from $x=2$ and using gradient descent with a learning rate of 0.1, what is the value of x after one update?

- ☒ 1.2
- ☐ 1.6
- ☐ 2.0
- ☐ 2.4

3. [单选] Alpha-Beta Pruning (0.5分)

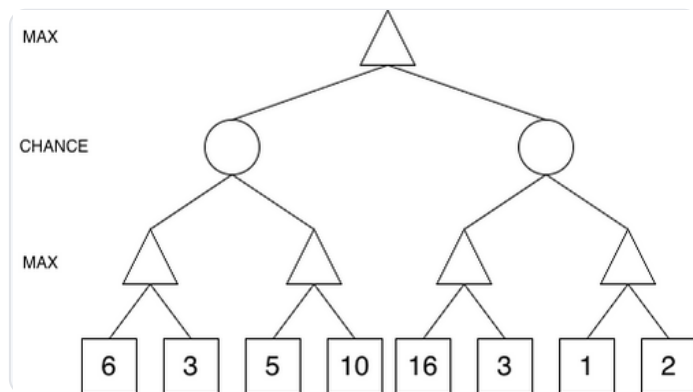
Consider the following game tree where MAX moves first. The leaf values are given. Which of the following edge will be pruned?



- ☐ a
 ☐ b
 ☐ c
 ☒ d

4. [单选] Expectimax (0.5分)

Given the following game tree, what is the value of the topmost MAX node? Suppose the chance node takes actions uniformly at random.



- ☒ 9
 ☐ 8
 ☐ 7
 ☐ 6

5. [单选] Genetic Algorithm (0.5分)

A population has four 4-bit chromosomes with their corresponding fitness values: (1010, 10), (1100, 25), (0110, 15), (0011, 5)

The two fittest individuals are selected as parents. Crossover is performed after the 2nd bit. What is the resulting offspring?

- ☒ 1110 and 0100
 ☐ 0110 and 1000
 ☐ 1011 and 0010
 ☐ 1111 and 0000